Kameti

Desi committee (BC) tracking app - project plan

1. Idea ka summary

Kameti ek web app hai jo kisi bhi BC/committee ka monthly record rakhti hai: kaun se member ne is mahine apna hissa diya, kisne nahi diya, aur is mahine payout kis ki baari hai. Jo member nahi deta usko email se reminder jaata hai. Admin aur member dono ke alag signup aur dashboard hain.

Cheez	Faisla
Payout order	Fixed (har member ka pehle se tay number)
Payment marking	Shuru mein manual (admin paid/unpaid mark karta hai); bank/SMS auto-detect baad mein
Users	Do signup: admin aur member (ek hi users table, role ke saath)
Admin to committees	Ek admin ki kitni bhi committees (one-to-many)
Member to committees	Ek member kitni bhi committees mein (many-to-many)
Member kaise judta hai	Dono: admin email se add kare, ya invite code/link
Reminder	Unpaid members ko email (pehle button se, baad mein automatic)
Stack	Vanilla JS + Node.js + Express + SQLite (baad mein PostgreSQL), Nodemailer, Railway/Vercel

2. Features

Area	Kya hoga
Accounts	Admin signup, member signup, login/logout, password hashing, role ke hisaab se dashboard
Committees	Naam, monthly amount, total members, start date; admin kitni bhi bana sakta hai
Membership	Member kitni bhi committees mein; har membership ka apna payout_turn (fixed order)
Cycles	Har mahine ka cycle; is cycle ka payout kis membership ko jaana hai
Payments	Admin har member ka paid/unpaid mark kare; source field (manual/email/sms); collected vs baaki total
Admin dashboard	Committee ka paid/unpaid view, unpaid ki list, remind button, is mahine ki payout baari
Member dashboard	Apni committees ki list, apna status, apni payout ki baari
Notifications	Unpaid ko email reminder (log ke saath); payment mark hone par app ke andar notification
Invites	Email se add, ya invite code/link se join (admin approval ke baad membership active)
Security	Har route pe permission check, input validation, rate limiting on login

3. Database design

Poora app in tables pe khada hai. Sabse zaroori baat: payout_turn aur payments ab user se nahi, membership se judte hain, kyunke ek hi insaan ek committee mein pehle number pe aur doosri mein aakhri pe ho sakta hai.

Table	Columns	Notes
users	`id, name, email (unique), password_hash, role (admin/member), created_at`	Admin aur member dono yahin
committees	`id, admin_id, name, monthly_amount, max_members, start_date, status`	admin_id to users (one-to-many)
memberships	`id, user_id, committee_id, payout_turn, status, joined_at`	Junction table; unique(user_id, committee_id)
cycles	`id, committee_id, cycle_no, due_date, payout_membership_id`	Har mahine ki ek row
payments	`id, membership_id, cycle_id, amount, status, paid_at, source`	Unique(membership_id, cycle_id); source = manual/email/sms
invites	`id, committee_id, email (nullable), code (unique), status, expires_at, max_uses, created_by`	Email invite aur code invite, dono yahin
email_log	`id, user_id, cycle_id, type, sent_at`	Spam rokne aur record ke liye
notifications	`id, user_id, message, seen, created_at`	App ke andar notifications

Relations

- **Admin to Committees:** one-to-many (committees.admin_id).
- **Members to Committees:** many-to-many, memberships table ke zariye.
- **Payment:** ek membership + ek cycle = ek payment row. 'Kisne nahi diya' = is cycle ki wo memberships jinki payment row nahi hai ya unpaid hai.

> **Paise ka rule:** amount hamesha integer mein store karo (rupees ya paisay), float mein kabhi nahi. Dates ka timezone (Pakistan, UTC+5) shuru se sochna, warna due date ka hisaab mahine ki seema pe galat aa sakta hai.

4. Member kaise judta hai (invites)

Tareeqa	Flow	Safety
Email se add	Admin email daalta hai. Agar user pehle se hai to usko accept ka option; nahi hai to signup link wali email jaati hai, signup ke baad wohi email match hone par committee judti hai.	Bina poochhe kisi ko committee mein na daalo
Invite code/link	Admin code banata hai, member /join/CODE pe jaake khud join karta hai.	Expiry (jaise 7 din), max uses, revoke button, crypto.randomBytes se code, admin approval

Dono mein common: committee full to nahi, user dobara to nahi judta (unique constraint), aur payout_turn admin baad mein edit kar sake.

5. Authorization (kaun kya kar sakta hai)

- Role signup route se tay hota hai (/signup/admin, /signup/member). Server kabhi client ke bheje role pe bharosa nahi karta.
- Admin sirf **apni** committees ke members, cycles aur payments badal sakta hai. Admin A, admin B ki committee nahi chhoo sakta.
- Member sirf **apni** memberships aur apne payments dekh sakta hai, kisi aur ke nahi.
- Yeh check middleware mein ho (jaise requireAdminOf(committeeId)), har route mein copy-paste nahi.

6. Build ka order (phases)

Schema shuru se hi many-to-many wala rakho, lekin features dheere dheere badhao taake momentum bana rahe.

Phase	Kya banana hai	Seekhna
0 - Practice	Express mein /members route, memory ke array ke saath (bina database)	Node basics, routing, express.json()
1 - Core	SQLite schema, admin signup/login, committee banana, members manually add karna	SQL, JOIN, parameterized queries, bcrypt, sessions
2 - Payments	Cycles, paid/unpaid toggle, collected vs baaki, admin dashboard	Transactions, GROUP BY, dates/timezone
3 - Reminders	Unpaid ko email (Nodemailer, Gmail App Password), email log, in-app notifications	SMTP, async flows
4 - Members	Member signup/login, member dashboard, permission middleware	Authorization, multi-role design
5 - Invites	Email invite pehle, phir invite code/link, admin approval	Tokens, expiry, unique constraints
6 - Deploy	GitHub, env variables, Railway pe deploy, persistent storage ya PostgreSQL	Deployment, dotenv
7 - Baad mein	node-cron automatic reminders, bank/SMS payment detect (confirm ke baad paid)	Cron, parsing, webhooks

> **Waqt ka andaza:** 3-4 ghante/hafta ke hisaab se manual MVP (phase 0-3) lagbhag 6-8 hafte. Member login, roles aur invites (phase 4-5) ke saath poora kaam shayad 10-12 hafte. Scope chhota rakho aur har phase ke baad kuch chalta hua dekho.

7. Seekhne ke topics

Topic	Kya aana chahiye
Node.js	Modules, npm/package.json, dotenv, event loop ka basic idea
HTTP / REST	GET/POST/PUT/DELETE, status codes, params vs query vs body, headers, cookies, CORS
Express	Routing, middleware chain, error handling, routes/controllers alag karna
SQL / database	Tables, foreign keys, one-to-many aur many-to-many, JOIN, GROUP BY, transactions, better-sqlite3
Authentication	bcrypt hashing, session vs JWT, httpOnly cookie
Security	Validation, XSS (innerHTML ki jagah textContent), SQL injection, rate limiting
Email	SMTP, Nodemailer, Gmail App Password
Frontend	Forms, fetch, DOM render, loading/error states
Deployment	Git/GitHub, env variables, Railway, SQLite ki file restart pe udne ka masla

8. Dhyan rakhne wali baatein

- **Bank/SMS auto-detect:** Pakistan mein aam logon ko personal account ke liye koi public bank API nahi milti. Email/SMS parse karna possible hai lekin har bank ka format alag hai aur badal sakta hai. Auto-detect ke baad bhi seedha paid mat karo, pehle 'payment detected, confirm?' dikhao.
- **Payment source field:** abhi se rakho (manual), taake baad mein auto-detect add karne par purane records alag pehchaane ja sakein.
- **Duplicate emails:** har reminder ka log rakho, aur automatic reminders se pehle manual button se test karo.
- **Passwords:** kabhi plain text mein nahi; reset-password flow baad mein add karna padega.
- **Sensitive data:** yeh logon ke paise ka record hai, to HTTPS (Railway/Vercel by default dete hain) aur har route pe permission check zaroori hai.

Agla qadam: Phase 0 se shuru karo (npm init, pehla Express route, /members memory ke array ke saath), phir usi ko SQLite se jodo.